

Beyond Likert Scales: Evaluating Conversational Affect Assessment from Free-Text Responses

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Abstract

Ecological Momentary Assessment (EMA) relies on repeated questionnaires such as the PANAS, which can cause survey fatigue and reduce data quality. We investigate whether a single spoken response to “How are you?” can alleviate part of this burden. To this end, we train a multi-task encoder that predicts both the presence and intensity of each of the 20 PANAS items. Training is conducted on synthetic, LLM-generated, and LLM-annotated German spoken responses, while validation relies on 120 real transcripts paired with self-reported PANAS ratings. Although item detection reaches a macro-F1 score of .70 on synthetic data, performance on real speech is unreliable: at a 0.5 decision threshold, the model covers only 18% of valid ratings, yielding a conditional MAE of 1.34 and a weighted κ of 0.29. Correlations at the scale level with self-reported PANAS are weak (PA: $r = .13$; NA: $r = .10$), and a confirmatory factor analysis on the predicted scores fails to converge. An auxiliary semantic loss shows no detectable effect. Finally, we discuss the challenges and advantages of verbalizing affect compared to rating it, and advocate for moving beyond rigid Likert-scale projections toward text-native metrics in future conversational agents.

1 Introduction

Quantifying latent construct remains one of the most fundamental yet persistent challenges in psychological research (Orth, 1983). Because affective states are latent constructs inaccessible to direct physical measurement (Cronbach and Meehl, 1955), psychometrics relies heavily on standardized self-report instruments (Krosnick, 1999; Likert, 1932), such as the widely validated Positive and Negative Affect Schedule (PANAS; Watson et al., 1988; Breyer and Bluemke, 2016). While standardized questionnaires offer established reliability and construct validity, they suffer from inherent structural limitations. Static items, rigid response scales,

and potential language shifts can introduce systemic bias (Krosnick, 1999). Furthermore, numeric scales reduce rich experiences to flat numbers, ignoring affective context. These methodological constraints are particularly magnified in Ecological Momentary Assessment (EMA; Shiffman et al., 2008). Designed to capture real-time fluctuations in naturalistic environments, EMA frequently imposes high survey burdens on participants. Over repeated prompts, this leads to severe questionnaire fatigue, central tendency bias, and careless or insufficient effort responding, which substantially degrades data quality (Eisele et al., 2022; Ulitzsch et al., 2025). While conversational interaction or free-text inputs could theoretically yield far richer contextual insight, systematically extracting psychometrically sound indicators from unstructured text has historically lacked standardized, scalable methodology.

Recent breakthroughs in Natural Language Processing (NLP) and Large Language Models (LLMs) present a novel paradigm to resolve the tension between qualitative richness and quantitative standardization (Demszky et al., 2023; Kjell et al., 2024). In particular, transformer architectures offer powerful capabilities for capturing the nuanced, highly contextual nature of human language (Kjell et al., 2024). By extracting contextualized semantic representations—for instance, via sentence embeddings or multi-layer feature representations (Reimers and Gurevych, 2019)—these models can capture subtle shifts in meaning across different usage contexts. Consequently, this enables models to infer psychological constructs directly from natural, unconstrained language (Hommel and Arslan, 2025; Kjell et al., 2019, 2022).

2 Related Work

Language-based prediction of psychological states. Natural language engineering has increas-

ingly been used to identify clinically and psychologically relevant states from language, including indicators of severe psychological distress, suicide risk, and depression. On social media, prior studies have predicted suicide risk from online posts (Mohammadi et al., 2019), detected multiple co-occurring emotions in tweets (Baziotis et al., 2018), and shown that the language of Facebook posts predicts later depression diagnoses recorded in medical records (Eichstaedt et al., 2018). These studies demonstrate that language carries clinically and psychologically relevant signal. They mostly rely on task-specific models trained on naturally occurring text, however, rather than on responses elicited in a structured assessment setting.

Free text versus rating scales. A more closely related line of research compares open-ended text responses with traditional rating scales. Kjell et al. (2019) showed that semantic measures derived from free-text answers converge moderately strongly with rating-scale scores ($r = 0.72$). Later work using transformer-based language models improved on this considerably: text and descriptive-word responses predicted established subjective well-being scales with accuracy approaching the theoretical upper limits set by scale reliability ($r = 0.85$, $N = 608$) (Kjell et al., 2022). Natural language also appears to carry more information than closed-ended formats. In a comparison of an open response to “How are you feeling?” with the PANAS, the free-text answers yielded roughly 4.8 times higher self-information (Diversity Index 366 vs. 77) (Kjell et al., 2024), so they distinguish between respondents better than the full rating-scale response.

To further bridge the gap between unstructured free text and formal psychometric assessment, we propose a multi-task learning (MTL) framework that derives multi-label PANAS information from a single German-language text response. A shared Transformer encoder feeds two primary heads: a binary presence head that detects which PANAS adjectives are referenced, and a regression head that estimates their Likert-scale intensity. Because the model is trained exclusively on synthetic, LLM-generated responses but is intended for empirical transcripts, it faces a domain shift. To mitigate this, we add a third, auxiliary head that predicts sentence-level cosine similarities between the response and the PANAS item embeddings, computed with Sentence-BERT (Reimers

and Gurevych, 2019).

3 Methods

3.1 Data Set Selection

Our task requires multi-label affect recognition from a single free-text statement in which a person describes their current state. A survey of publicly available sentiment and affect data sets revealed two mismatches with this setting. First, most corpora consist of product or media reviews rather than self-reports of affective state. Second, the few corpora with affective statements (e.g., online chat comments or Reddit dialogues) provide only a single emotion label per text. Moreover, the PANAS measures state affect on two dimensions and is therefore not equivalent to discrete emotion categories. Training on only loosely related data risks introducing systematic noise and shifting the model away from the target construct.

We therefore decided to compile a data set of our own from two sources: real self-reports and synthetic texts. Real data promise high validity but are costly to collect at scale. Synthetic data are cheap to generate, but the quality of the texts and the accuracy of their PANAS labels have to be established. We used the synthetic texts for training, validation, and testing, and the smaller real-world set exclusively as an independent external validation set.

3.2 Real-World Data

Real-world data were collected with an online survey on SoSci Survey. Participants first entered a pseudonym composed of the first two letters of their last name, the first letter of their first name, the time (HHMM) and the date (DDMMYY), e.g., FEA2004110926. They then recorded a spoken description of their current affective state and subsequently completed the German PANAS (Breyer and Bluemke, 2016). The recorded audio file was saved on SoSci Survey and later locally transcribed with Whisper (Radford et al., 2023). The pseudonym links all recordings of one person, which allowed us to account for repeated measurements in the analysis (participant-level bootstrap). Participation was voluntary and repeated participation was possible. Participants were recruited through the authors’ personal networks. In total, we obtained 120 transcribed responses from 36 unique participants. Of the 2,400 possible PANAS ratings, 2,180 were valid and 220 missing; no value fell outside 1–5.

These data were used neither for training nor for checkpoint selection. Because the questionnaire forces a rating for every item, a rating does not indicate whether an item was actually mentioned.

3.3 Synthetic Data

The synthetic data were generated and labeled with LLMs (all prompts are available in the repository). The prompts pursued three goals: (i) natural-sounding, spontaneous speech, (ii) labels that two independent raters can reproduce, and (iii) a clear distinction between affects that are expressed explicitly or implicitly and affects for which the text provides no information.

Generation and labeling. Claude Sonnet 5 (reasoning effort: high) generated 1,000 texts in response to the question “Wie geht es dir im Allgemeinen gerade?” (Prompt 1). A subset of 350 texts was rated independently by Claude Sonnet 5 (high) and DeepSeek-V4.1-Flash on all 20 PANAS items (1–5). In a first try, we prompted the models to generate matching PANAS scores for all items that were not mentioned in the texts with natural metric variance. Since reliability ratings were poor, we switched our prompting strategy. Consequently, items without sufficient textual evidence were coded as 0 (“not assessable”) instead of being estimated (See Prompt 2 in our repository), so that the ratings reflect only what a text supports.

Inter-rater agreement. Agreement improved with this approach but remained low for some items, partly because certain items were rarely expressed in the original texts. We therefore generated additional texts that specifically target these items (see Prompt 3), resulting in a total of 650 rated texts. Table 1 reports Cohen’s κ (Cohen, 1968) for each item.

Following the benchmarks of Landis and Koch (1977), weighted κ indicated substantial to almost perfect agreement for 18 of 20 items, with a mean of 0.77. Agreement was moderate for *aktiv* (0.57) and only slight for *stark* (0.13). We therefore interpret them with caution. Since the labels of the synthetic data are model-generated and no human gold standard exists, inter-rater agreement establishes consistency, not validity; the latter is assessed on the real-world data with self-reported PANAS scores. Based on these results, the remaining texts were labeled by DeepSeek-V4.1-Flash alone.

PANAS item	Unw. κ	Weighted κ
aktiv	0.273	0.569
bekümmert	0.540	0.814
interessiert	0.709	0.942
freudig erregt	0.519	0.662
verärgert	0.749	0.977
stark	0.052	0.132
schuldig	0.333	0.666
erschrocken	0.399	0.889
feindselig	0.308	0.670
angeregt	0.291	0.627
stolz	0.496	0.867
gereizt	0.670	0.795
begeistert	0.458	0.671
beschämt	0.971	0.997
wach	0.331	0.920
nervös	0.646	0.908
entschlossen	0.731	0.917
aufmerksam	0.687	0.812
durcheinander	0.521	0.704
ängstlich	0.561	0.803
Mean	0.512	0.767

Table 1: Inter-rater agreement (Cohen’s κ) between the two LLM raters for each PANAS item. Weighted κ uses quadratic weights.

The final synthetic set comprises 1,400 labeled texts. On average, 7.1% of all item labels are positive (value > 0). The texts were randomly split into training (896), validation (224), and test (280) samples (seed 42).

3.4 Model Architecture

To simultaneously tackle item detection and affective intensity estimation in free-text statements, we propose a multi-task learning architecture based on PyTorch. The core objective is twofold: (1) identifying whether a participant implicitly or explicitly references any of the 20 PANAS affective states, and (2) predicting the corresponding intensity on a continuous Likert-scale equivalent (1–5).

We employ a shared encoder-only backbone based on bert-base-german-cased (Devlin et al., 2019) and distillbert (Sanh et al., 2019) to compare the impact of different amount of layers in our encoding structure. The final hidden state of the [CLS] token, after applying dropout ($p = 0.1$), is routed into three parallel linear task heads, each projecting to a 20-dimensional output space (one value per PANAS item):

1. A *Presence Head* outputting raw logits for binary item detection;
2. A *Likert Head* providing continuous intensity estimates;

3. An *Auxiliary Cosine Head* serving as a semantic regularizer.

Presence Head Detecting mentioned PANAS items presents a severe class imbalance challenge: in open-ended participant statements, individual utterances typically cover only a small subset of the 20 PANAS dimensions. The same issue arises in the synthetic training data, since the free text sample was created independent of later PANAS scoring. To prevent hallucinated annotations, unmentioned affective states were strictly assigned zero labels (0), rendering zero instances the vast majority (> 90%).

We address this imbalance through two complementary mechanisms:

- (a) **Dynamic Positive Class Weighting:** Item-specific positive weights $w_i = N_{\text{neg},i}/N_{\text{pos},i}$ are incorporated into the Binary Cross-Entropy loss, capped at a maximum ratio of 20.0 to ensure numerical stability across highly sparse items (e.g., *aktiv*, *beschämt*, *stolz*, *schuldig*, *stark*, *aktiv*).
- (b) **Loss Scaling:** The presence loss multiplier is set to $\lambda_P = 3.0$ to prioritize detection performance within the joint gradient update.

For real-world evaluation datasets lacking explicit ground-truth annotations for unmentioned items, presence classification is learned exclusively on the synthetic split. Intensity predictions are subsequently evaluated conditionally on predicted active items.

Likert Head Affective intensity can only be meaningfully quantified for items that are explicitly or implicitly present in the text. Consequently, the Likert regression head is trained conditionally, masking out instances where no affect was detected. In the synthetic training corpus, this restricts parameter optimization to the subset of active items ($\approx 7.1\%$ of total item instances). The head consists of an unconstrained linear projection mapping the dimensional [CLS] representation to continuous intensity values, optimized via Mean Squared Error (MSE) against the target range [1.0, 5.0].

Auxiliary Cosine Head To enrich the shared encoder with domain-informed semantic alignment signals (Ruder, 2017), we incorporate an auxiliary regression task using pre-computed Sentence-BERT embeddings (Reimers and Gurevych,

2019). Prior to fine-tuning, transcripts are segmented into sentences at sentence boundaries. Dense L_2 -normalized vector representations are extracted for both individual sentences and canonical PANAS adjective descriptors using paraphrase-multilingual-MiniLM-L12-v2 (Reimers and Gurevych, 2020).

For each PANAS item i , a continuous target value c_i is derived as the maximum cosine similarity across all S sentences in the utterance:

$$c_i = \max_{s \in \{1, \dots, S\}} \cos(\mathbf{e}_s, \mathbf{e}_i). \quad (1)$$

By supervising the linear auxiliary head to predict these continuous similarity scores across all items (unmasked), the shared backbone is regularized toward a continuous, semantic representation of affect.

3.5 Training Objective

The model is optimized via a combined weighted loss formulation:

$$\mathcal{L}_{\text{total}} = \lambda_P \mathcal{L}_{\text{presence}} + \lambda_L \mathcal{L}_{\text{Likert}} + \lambda_C \mathcal{L}_{\text{cosine}}, \quad (2)$$

where $\lambda_P = 3.0$, $\lambda_L = 1.0$, and $\lambda_C \in \{0.0, 0.5\}$.

The loss components are defined as follows:

- $\mathcal{L}_{\text{presence}}$ is a binary cross-entropy loss with logits ('BCEWithLogitsLoss'), incorporating the item-wise positive class weights $w_i \in [1.0, 20.0]$ computed on the training split.
- $\mathcal{L}_{\text{Likert}}$ is an item-masked regression loss computed strictly over active items:

$$\mathcal{L}_{\text{Likert}} = \frac{\sum_i m_i \cdot \ell(y_{\text{pred},i}, y_{\text{true},i})}{\sum_i m_i + \epsilon}, \quad (3)$$

where $m_i = \mathbb{I}[y_i > 0]$ denotes the binary presence mask, $\ell(\cdot)$ represents the regression metric (MSE), and $\epsilon = 10^{-8}$ prevents division by zero.

- $\mathcal{L}_{\text{cosine}}$ is an unmasked Mean Squared Error evaluated across all 20 items, penalizing deviations from the Sentence-BERT reference similarities.

Optimization is performed using AdamW (Loshchilov and Hutter, 2019) with an initial learning rate of 2×10^{-5} , a linear warmup over 10% of total training steps, followed by linear decay. Gradient norms are clipped at 1.0. Models are trained for 5 epochs with a batch size of 8 and a maximum sequence length of 256 tokens using dynamic padding.

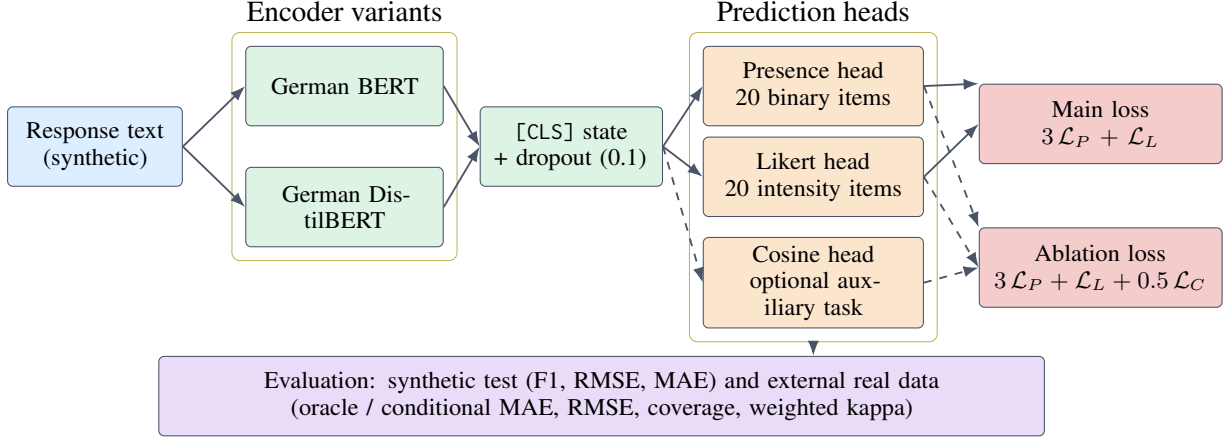


Figure 1: Architecture of the multi-task model. Response text is encoded using either German BERT or German DistilBERT, followed by separate linear heads for presence and Likert-scale intensity prediction. An auxiliary cosine head is included only in the ablation setting. $\mathcal{L}_L$ is computed only over items present in the text; $\lambda_C \in \{0, 0.5\}$.

3.6 Ablation and Model Selection

To systematically evaluate the impact of the semantic auxiliary regularizer, we conduct an ablation study comparing models trained with $\lambda_C = 0.5$ against a baseline omitting the auxiliary head ($\lambda_C = 0.0$).

To ensure statistical robustness, all configurations are trained across three independent random seeds (42, 43, 44), controlling for parameter initialization and mini-batch sequencing while maintaining identical data splits. Model selection is governed strictly by the primary validation loss:

$$\mathcal{L}_{\text{main}} = 3\mathcal{L}_{\text{presence}} + \mathcal{L}_{\text{Likert}} \quad (4)$$

Excluding $\mathcal{L}_{\text{cosine}}$ from the selection metric ensures unbiased checkpoint selection across both experimental arms. Final held-out synthetic test evaluations and external real-data inference are conducted exclusively using the optimal validation checkpoint.

Oracle and conditional approach To isolate the quality of the Likert head, we first evaluated all valid ratings irrespective of the presence probability outcome (oracle setting, coverage = 1). In the *conditional* setting, only valid ratings whose presence probability reaches a threshold are scored, which approximates the intended pipeline but depends on the presence head. The primary threshold was fixed a priori at 0.5; thresholds of 0.3, 0.7, and 0.9 were examined as a sensitivity analysis of the trade-off between error and coverage (share of valid ratings evaluated). Likert predictions were clipped to $[1, 5]$. We report MAE, RMSE, and the quadratic

weighted kappa (κ), computed both over all evaluated ratings (pooled) and per item (mean item-wise κ).

As baselines we used the item-wise mean and mode of the Likert values in the synthetic training split (computed over items with detected affect). Since ratings within a participant are not independent, to account for seed-to-seed variance across multiple runs, metrics and confidence intervals were computed across the aggregated predictions of all seeds, capturing both within-run uncertainty and cross-run variability.

4 Results

We analysed our data from two perspectives: First, on item level, where we looked into Presence Score (was the item correctly detected by our model?) and Likert Score (For all detected items: Was the likert estimate close to the true likert value?). Second, on scale level: PANAS is traditionally not used on item level, but a positive and negative affect score is computed by averaging over the PANAS scores in the positive affect items and negative affect items separately.

4.1 Item Level

Presence Score To evaluate the model’s ability to detect whether a specific PANAS affective state is expressed in a given text (Presence Score), we report macro-F1 (at threshold 0.5) alongside overall accuracy.

While accuracy provides a baseline metric, it is heavily skewed by the significant class imbalance in our dataset (over 90% zero-labels across items).

Metric	$\lambda_C = 0$	$\lambda_C = 0.5$
BERT F1	.703	.702
BERT Accuracy	.963	.963
DistilBERT F1	.676	.673
DistilBERT Accuracy	.955	.954

Table 2: Synthetic test set performance, averaged over three seeds.

Consequently, the macro-F1 score serves as our primary evaluation metric, as it provides a balanced assessment of Precision and Recall regardless of class frequency.

Across all experiments, introducing the auxiliary cosine loss ($\lambda_C = 0.5$) yielded virtually identical results compared to the baseline without cosine regularization ($\lambda_C = 0$). For BERT, macro-F1 scores remained essentially stable at .702 vs. .703, while DistilBERT showed a minor drop from .676 to .673. Accuracy remained virtually unchanged at .963 for BERT and .955 for DistilBERT. Overall, full BERT outperformed DistilBERT by approximately 2.7 to 3 percentage points in macro-F1. The evaluation is limited exclusively to synthetic test data. Because our real-world dataset lacks dense item-level ground-truth annotations, it is difficult to reliably estimate how well these detection capabilities generalize to real-world expressions. Synthetic datasets often exhibit lower linguistic diversity and more predictable affective patterns than real-world text, meaning these metrics likely represent an optimistic upper bound rather than realistic out-of-domain performance in spite of our efforts including the auxiliary cosine loss.

The item-wise F1 scores (for our best model: bert_ohne_cosine | Seed=44 | Epoch=5) show substantial variation across affect items. "feindselig" achieves a high F1 score of 0.9524, which may partly reflect class imbalance, as the model can often default to predicting absence for sparse items. In contrast, schuldig (Item 6) and erschrocken (Item 7) yield $F1 = 0.0$, indicating insufficient positive support and difficulties in learning these rare classes.

The oracle approach established a strong upper bound, significantly outperforming naive mean and mode baselines across all error metrics (MAE = 1.1874, Within-1 Acc = 66.59%)

Likert Scale These oracle results indicate that the model can learn intensity-related patterns from the synthetic training distribution when item pres-

Encoder	Cosine	PA	NA
BERT	With	0.125 ± 0.038	0.075 ± 0.008
BERT	Without	0.131 ± 0.041	0.079 ± 0.006
DistilBERT	With	0.127 ± 0.024	0.101 ± 0.009
DistilBERT	Without	0.129 ± 0.022	0.102 ± 0.010

Table 3: Pearson correlations between real and predicted PANAS scale scores, reported as mean $\pm$ SD over three seeds.

ence is assumed to be known (MAE = 1.1874, Within-1 Acc = 66.59%). Furthermore, while DistilBERT degraded under extreme sparsity ($\kappa = 0.0709$ at $\kappa=0.90$), BERT maintained high calibration and superior exact accuracy (32.65%). Unfortunately across all threshold levels, auxiliary cosine regularization provided no measurable benefit.

Evaluating the LIKERT task itemwise: The low MAE for "schuldig" (0.3130) and "erschrocken" (0.4730) is partly explained by the model consistently predicting zero, which is often close to the true value due to the sparsity of these items. In contrast, frequently occurring items with greater intensity variation are harder to predict accurately. Overall, negative affect is estimated more precisely than positive affect, with MAE values below 1.0 for "verärger" (0.9822), "feindselig" (0.6322), "gereizt" (0.7451), and "beschämt" (0.6389).

4.2 Scale Level

Correlations between Affective Scales Since the PANAS is primarily analyzed at the scale level (Positive Affect, PA, and Negative Affect, NA), we evaluated our models on these aggregated scores. For each participant, scale scores were computed as the mean of the ten PA and ten NA items for both real ($PA_{\text{real}}, NA_{\text{real}}$) and predicted scores ($PA_{\text{pred}}, NA_{\text{pred}}$).

Table 3 shows the Pearson correlations averaged across three seeds. Overall, correlations were weak across all configurations. For PA, models achieved comparable performance ($r = .125-.131$), with DistilBERT showing lower variance across seeds (SD $\approx .02$) than BERT (SD $\approx .04$). For NA, correlations were lower ($r = .075-.102$), with DistilBERT slightly outperforming BERT. The auxiliary cosine loss provided no benefit at the scale level.

Confirmatory Factor Analysis To evaluate structural validity, a two-factor CFA was fitted on real and predicted scores. On real data ($N = 109$), all items loaded significantly on their target factors (PA: .60-.83; NA: .41-.79; factor correlation

$r = -.56$), yet with poor to moderate global fit (CFI = .743, RMSEA = .132). For predicted scores (BERT baseline, seed 42), the CFA failed to converge due to a non-positive-definite covariance matrix. Low variance and near-zero loadings in predicted items (e.g., *wach*, *beschämt*) indicate that the model fails to reproduce the empirical item-covariance structure, aligning with the low scale-level correlations.

5 Discussion

The empirical evaluation reveals a critical divergence between item-level detection performance and scale-level psychometric validity when predicting PANAS affective states from unstructured text. While BERT achieved a macro-F1 of .703 on synthetic data, these scores are inflated by severe class imbalance: over 90% of labels were zeros. Rare negative states like *schuldig* and *erschrocken* yielded F1 = 0.00, and low MAE for these items reflects regression to the mean rather than predictive precision. This pattern is consistent with known difficulties in imbalanced multi-label classification (Lin et al., 2025).

These item-level distortions compound at the scale level. Correlations between real and predicted PANAS scores were weak (PA: $r = .125-.131$; NA: $r = .075-.102$), and auxiliary cosine regularization provided no benefit. The confirmatory factor analysis failed to converge due to near-zero variance in several predicted items. The predicted scores did not permit a stable CFA, primarily because several items exhibited near-zero variance. Thus, the model output did not reproduce a covariance structure suitable for the intended two-factor measurement model. ($r = -.56$ between PA and NA in real data). This tempers enthusiasm for natural language evaluation (NLE). While Kjell et al. (2022) reported $r = .85$ for well-being scales, such correlations do not automatically generalize to item-dense inventories like the PANAS. The PANAS itself showed only moderate fit on our real data (CFI = .743), consistent with some validation studies (Crawford and Henry, 2004).

Several factors explain these boundary conditions. Real-world free text introduces unstandardized lengths, class sparsity, and social desirability bias, complicating supervised alignment. EMA research has long documented that longer questionnaires increase burden and careless responding (Eisele et al., 2022). Synthetic data create an op-

timistic upper bound: the F1 of .703 on synthetic data contrasts with only 18% coverage on real data at $\tau = 0.5$. Synthetic data can help but require careful validation (Cabrera Lozoya et al., 2025). A single open-ended prompt cannot reliably evoke all 20 PANAS states; multi-turn dialogic probing is needed. Conversational agents in EMA show promise for adherence (Ahmadi et al., 2025). Finally, restrictive presence gating improved calibration (QWK = 0.2911) but collapsed coverage to < 1.6% and introduced positive selection bias.

Despite these constraints, qualitative analysis showed that free text carries contextual information that Likert scales miss. The model detected implicit activation in phrases like “*frisch und aufgewacht*.” This aligns with work showing free-text responses are perceived as more precise than rating scales (Kjell et al., 2023). However, language models also inherit the psychometric vulnerabilities of their training data, including class imbalance and skewed distributions. LLM-generated data may perpetuate or amplify such biases (Li et al., 2025).

The path forward requires moving away from rigid Likert projection toward text-native metrics. Promising alternatives include diversity indices, semantic network analyses, and theory-driven causal formulations. The CONCORD framework, for example, recovers individualized causal structures from natural conversation (Hertz-Palmor et al., 2025). Semantic embeddings can also address taxonomic incommensurability in measurement (Wulff and Mata, 2025). Concrete steps include multi-turn agents, more realistic synthetic data, distributional losses for uncertainty quantification (Amini et al., 2020), and evaluation on larger, densely annotated real-world datasets.

In sum, replacing rigid questionnaires with conversational agents is promising but immature. Our work shows both potential and limits: item-level detection can succeed on synthetic data, but generalization and scale-level validity remain open challenges. Language carries richer information than Likert scales, but a direct projection from free text onto an existing Likert instrument is not sufficient under the tested setup. From our point of view, supervised affect detection currently remains bottlenecked by the availability of high-quality real-world annotations.

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Additional Information and Material

Please visit: [audio2PANAS on github.com](https://github.com/audio2PANAS)

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